

Salman Khalid Alwajaan

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PROFILE

Electrical Engineer focused on renewable and electrical energy systems, NEBOSH-certified compliance and risk assessment skills. Passionate about renewable energy integration, automation, and improving power system performance.

EDUCATION WORK

Bachelor of Engineering (B.E.) - Renewable Energy Engineering University of Hail, Hail
Saudi Arabia, Mar 2019 – Jan 2026

- Senior Design Project – Stand Alone Solar Water Heating System with PCM Storage tank : Senior Design Project, at UOH Designed and built an off-grid solar water heater with PCM thermal storage, achieving 63% collector efficiency and heating 30L of water/day from 23°C to 80°C; PCM storage retained heat 9°C longer than a conventional tank after 12 hours.

PROJECT

- SolarPV Studio: A local web application for solar photovoltaic system simulation: A desktop app you run locally that simulates how a solar panel system would perform and pays for itself, sizing, energy output, and cost/return, with a report generated automatically.
- Tuwaiq Mechatronics Project: Simon Says, an embedded interactive device built around an ESP32, combining gesture sensing, multi-color lighting, audio feedback, and an onboard display into a single real-time, escalating challenge system, designed and built with a team at Tuwaiq Academy.
- SAM-6 Solar UAV: solar fixed wing, powered entirely by solar panels on its wings, designed to stay in the air for a very long time without needing to land and recharge

Technical Skills

- Software: Homer Pro, Ansys, PVsyst, TRNSYS, Power BI, SolidWorks, Kicad, EasyEDA.
- Government & HR Platforms: ERP SYSTEM, Absher, GOSI, Qiwa, Mudad, Muqem, MHRSD
- Embedded Systems: (Arduino, ESP32, STEM), Sensor Integration, Soldering (IPS Standards), Communications Protocols, 3D Molding.
- Programming: Python, C, HTML.
- Soft Skills: Communication, Teamwork, Adaptability, Fast Learning, Excellent analytical and problem-solving skills. Effective communication and teamwork abilities.

WORK EXPERINCE

Safety Intern – HSE Department
Barq International Holding Company – Hail, Saudi Arabia
June 2024 – February 2025 (Training Program)

- Supported ISO 45001 implementation through site inspections, toolbox talks, and internal safety audits.
- Assisted with incident investigations and digital safety reporting, and helped build a local-first QHSE ERP system used for compliance tracking against Saudi electrical and SEC standards.

Certifications

- NEBOSH International General Certificate in Occupational Health and Safety (IGC), 2025
- Foundations of Mechatronics, Tuwaiq Academy, 2026

Languages

- Arabic: Native • English: Fluent